

For the first few hundred years of classical Greek civilization, scientific thought was not unlike that of earlier periods. Just as in Mesopotamia and Egypt, practical knowledge was used to help make life easier. Some individuals acquired skills in mathematics and astronomical observation, but they were used for practical purposes. Mathematics, therefore, was used in commerce and construction, and knowledge of astronomy in timekeeping.

When it came to accounting for natural phenomena, however, ancient civilizations tended to be far less practical. They relied unquestioningly on fanciful mythological explanations for everyday events. Myths involving supernatural gods were not tested—or even questioned.



Praising the gods

This clay tablet was left at a temple, and was dedicated to god of healing, Asclepius, from someone who believed their foot to have been healed through traditional rather than scientific medicine.

After about 600 BCE, however, some began to question these ideas and wonder about the world around them.

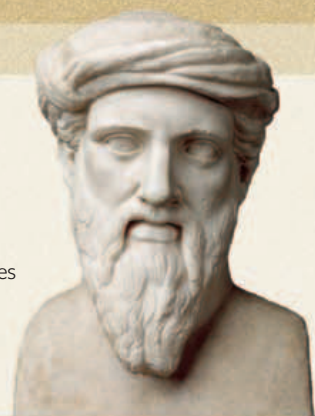
Natural philosophers

These thinkers, or “natural philosophers,” employed logic and reason to question the accepted myths. Thales of Miletus (c. 624–c. 526 BCE) is often referred to as the “father of science.” Like many of the Greek scientific thinkers, Thales pondered the nature of matter. He suggested that all solids, liquids, and gases are ultimately made of water. This makes some sense—water is one of

GREEK PHILOSOPHER (c. 580–c. 500 BCE)

PYTHAGORAS

Pythagoras is best known for the mathematical law relating the sides of right-angled triangles: “the square of the hypotenuse of a right-angled triangle is equal to the sum of the squares of the other sides.” In fact, the theorem was known long before Pythagoras, and there is no evidence that he even used it, or whether he even existed. His reputation for mathematical excellence comes from later Greek writers, particularly Aristotle. Modern academic study of Pythagoras’ ideas shows that they were more religious than scientific. Pythagoras and his followers saw mystical relevance in mathematics.



(born c. 480 BCE) and his student Democritus (born c. 460 BCE) suggested that matter is made up of tiny, indivisible particles separated by empty space. Their ideas are similar to modern atomic theory.

All of these early thinkers grappled with philosophical concepts, such as change, infinity, and existence versus nonexistence (see pp. 130–31). It is not important whether the theories were

mathematics in formulating their theories. Mathematics can be divided into geometry—the study of the relative position and size of objects, including Earth—and arithmetic—the study of numbers. Geometry had been used to help astronomers and architects, while arithmetic formed the basis of commerce. The natural philosophers used mathematics as a way of seeking truth.

Triumphs of Greek Science

Science is an attempt to understand and explain the world around us. The modern scientific method—a combination of observation, hypothesis, experiment, and theory—was established in the 17th century. But its roots lie with the ancient Greeks, who were among the first to think scientifically and search for plausible answers to life’s mysteries through logic and observation.

BEFORE

Some of the scientific ideas of the Greeks had been considered by earlier civilizations.

RATIONAL THOUGHT

In ancient India and China, philosophers thought about the world rationally, explaining causes of events by “laws,” rather than myth.

EGYPTIAN ASTRONOMY

In Mesopotamia, the Babylonians and Egyptians had writing systems and calendars, and managed **impressive construction projects** << 70–71 that required **exact measurements**.



EGYPTIAN CONSTELLATIONS

the few substances that we can observe directly changing between these three states. Another philosopher, Anaximenes (585–525 BCE), suggested that air was the “fundamental” substance, while Heraclitus (535–475 BCE) thought fire might fit the role. Inevitably, a more sophisticated and believable theory emerged, drawing on those that had gone before—the theory of the four elements, put forward by Empedocles (490–430 BCE). This theory explained many common phenomena in terms of the movement and interaction of air, earth, fire, and water.

Toward the end of the 5th century BCE, a new approach emerged. Leucippus

right or wrong, or even plausible, but that the people who constructed them were thinking rationally and philosophically, and were making observations to satisfy human curiosity. That can be seen as the true essence of science, and it had been missing from almost all earlier civilizations.

Philosophers in the ancient civilizations of India and China also employed reason and observation in their attempts to understand the world. They, too, formulated convincing

theories similar to the theory of the four elements and atomic theory. In many cases, they made key discoveries before the Greeks. But the lineage of Western scientific thinking leads directly back to ancient Greece.

Science counts

Thales, and many of the other natural philosophers, importantly used

Some philosophers derived elegant mathematical proofs that provided insight into the nature of the world. One of them, Pythagoras (see above), went one step farther, suggesting not only that numbers describe reality, but that the world is literally made of numbers. Pythagoras was fascinated by the mathematical relationships between everything, including musical notes. The connection between mathematics and reality is still a central theme in science today.

Mathematics was also crucial to the scientific ideas of Plato (c. 427–c. 347 BCE), better known for his writings on ethics and politics. He suggested that each element is made of atoms with a particular idealized geometric shape. Plato supposed that the real world was an imperfect reflection of an ideal, “theoretical” and mathematically perfect world.

The Greek philosopher who had the greatest influence on the history of science was

Euclid’s theories

Euclid of Alexandria, a Greek mathematician (c. 325–c. 265 BCE) living in Egypt, wrote a geometry book called *The Elements* (a folio of which is shown here). His ideas on geometry and number theory remain key to mathematics today.

